A Level • Cambridge (CIE) • Chemistry

41 mins

22 questions

Multiple Choice Practice

Topical Practice MCQ

11. Group 17

Carefully selected past paper questions by topic.

19 U, V and W represent different halogens. The table shows the results of nine experiments in which aqueous solutions of U_2, V_2 and W_2 were separately added to separate aqueous solutions containing U^-, V^- and W^- ions.

	U^-(aq)	V^-(aq)	W^-(aq)
U_2(aq)	no reaction	no reaction	no reaction
V_2(aq)	U_2 formed	no reaction	W_2 formed
W_2(aq)	U_2 formed	no reaction	no reaction

Which row contains the ions U^-, V^- and W^- in order of their **decreasing** strength as reducing agents?

	strongest	$\longrightarrow$	weakest
A	U^-	V^-	W^-
B	U^-	W^-	V^-
C	V^-	W^-	U^-
D	W^-	U^-	V^-

22 Compound Q dissolves in water. Q(aq) does **not** react with dilute sulfuric acid.

Q(aq) forms a precipitate when aqueous silver nitrate is added. This precipitate is partially soluble in aqueous ammonia.

What could be compound Q?

A barium bromide

B barium iodide

C magnesium bromide

D magnesium iodide

21 In reaction 1, concentrated sulfuric acid is added to potassium chloride and the fumes produced are bubbled into aqueous potassium iodide solution.

In reaction 2, potassium chloride is dissolved in aqueous ammonia and this is then added to aqueous silver nitrate.

What are the observations for reactions 1 and 2?

	observation for reaction 1	observation for reaction 2
A	brown solution	colourless solution
B	brown solution	white precipitate
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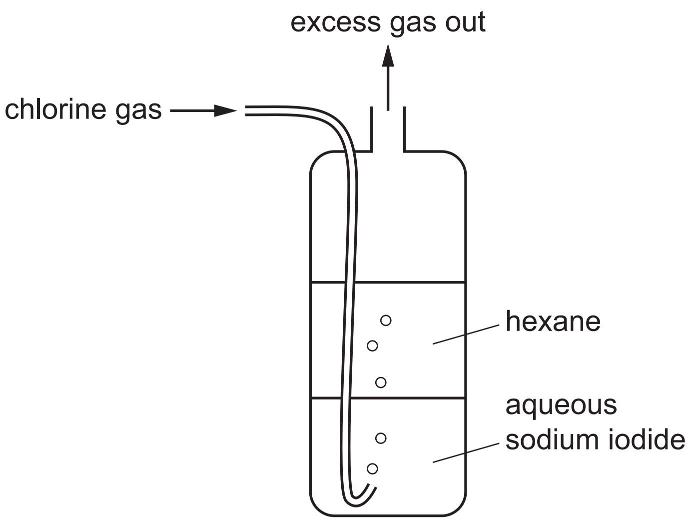
Question 5

23 An excess of chlorine was bubbled into 100 cm^3 of hot 6.0 mol dm^{-3} sodium hydroxide.

How many moles of sodium chloride would be produced in the reaction?

A 0.30 **B** 0.50 **C** 0.60 **D** 0.72

22 Chlorine is bubbled through a cylinder containing aqueous sodium iodide and an immiscible layer of hexane.



As the bubbles pass through the cylinder, what is observed in the lower and upper layers?

	lower aqueous layer	upper hexane layer
A	colourless solution becomes brown	colourless liquid becomes coloured
B	colourless solution becomes brown	colourless liquid is unchanged
C	brown solution becomes colourless	colourless liquid becomes coloured
D	brown solution becomes colourless	colourless liquid is unchanged

16 In a series of nine experiments, to test the reactivity of the halogens, an aqueous solution of each halogen is added to an equal volume of an aqueous solution containing halide ions, as shown in the table.

halogen solution	halide solution		
	sodium chloride (aq)	sodium bromide (aq)	sodium iodide (aq)
chlorine (aq)	experiment 1	experiment 2	experiment 3
bromine (aq)	experiment 4	experiment 5	experiment 6
iodine (aq)	experiment 7	experiment 8	experiment 9

The nine resulting mixtures are then shaken separately with an equal volume of hexane. The nine tubes are left to stand so that the aqueous and organic solvents separate into layers.

How many test-tubes contain a purple upper hexane layer?

A 1 **B** 2 **C** 3 **D** 5

14 $AgNO_3$(aq) is added to a solution of a halide ion, X^-(aq), and aqueous ammonia is then added.

The ionic equations for the two reactions that occur are shown.

$$Ag^+(aq) + X^-(aq) \rightleftharpoons AgX(s) \qquad \text{equilibrium 1}$$

$$Ag^+(aq) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq) \qquad \text{equilibrium 2}$$

Which statement is correct?

A The position of equilibrium 1 lies to the left when $X^- = I^-$.

B Increasing the concentration of ammonia causes the position of equilibrium 1 to move to the left.

C K_c for equilibrium 2 is larger when $X^- = Cl^-$ than when $X^- = I^-$.

D Equilibrium 2 is a redox reaction.

17 The reaction of bromine with warm NaOH(aq) produces products with the same oxidation numbers, in the same ratios, as the reaction of chlorine with hot NaOH(aq).

In one reaction between bromine and warm NaOH(aq), 30.2 g of a product containing sodium, bromine and oxygen is produced.

Which mass of NaOH has reacted?

A 8.00 g **B** 10.2 g **C** 20.3 g **D** 48.0 g

17 In an experiment, 0.600 mol of chlorine gas, Cl_2, is reacted with an excess of hot aqueous sodium hydroxide. One of the products is $NaClO_3$.

Which mass of $NaClO_3$ is formed?

A 21.3 g **B** 44.7 g **C** 63.9 g **D** 128 g

19 After black and white photographic film has been developed, unreacted silver bromide is removed by reaction with sodium thiosulfate.

$$AgBr + 2Na_2S_2O_3 \rightarrow 4Na^+ + Br^- + [Ag(S_2O_3)_2]^{3-}$$

What is the function of the thiosulfate ion?

A to make the silver ions soluble

B to oxidise the silver ions

C to reduce the bromine

D to reduce the silver ions

18 An excess of chlorine gas, Cl_2, is passed through $60\,cm^3$ of cold aqueous $0.1\,mol\,dm^{-3}$ sodium hydroxide. In a separate experiment an excess of chlorine gas is passed through $60\,cm^3$ of hot aqueous $0.1\,mol\,dm^{-3}$ sodium hydroxide until no further reaction takes place.

How much **more** sodium chloride will be produced by the reaction with hot NaOH than with cold NaOH?

A 0.002 moles

B 0.003 moles

C 0.005 moles

D 0.006 moles

18 In a series of nine experiments to test the reactivity of the halogens, an aqueous solution of each halogen was added to an equal volume of an aqueous solution containing halide ions as shown in the table below.

solution	sodium chloride (aq)	sodium bromide (aq)	sodium iodide (aq)
chlorine (aq)	experiment 1	experiment 2	experiment 3
bromine (aq)	experiment 4	experiment 5	experiment 6
iodine (aq)	experiment 7	experiment 8	experiment 9

The nine resulting mixtures were then shaken with hexane. The nine tubes were corked and left to stand so that the aqueous and organic solvents could separate into layers.

How many test-tubes contained a purple upper layer?

A 1 **B** 2 **C** 3 **D** 5

9 Brine is concentrated aqueous sodium chloride. Brine is electrolysed in a diaphragm cell.

What is the purpose of the diaphragm?

A to prevent Cl_2 reacting with H_2

B to prevent HCl reacting with Na

C to prevent NaOH reacting with Cl_2

D to prevent NaOH reacting with HCl

17 *Use of the Data Booklet is relevant to this question.*

In an experiment, 0.6 mol of chlorine gas, Cl_2, is reacted with an excess of hot aqueous sodium hydroxide. One of the products is a compound of sodium, oxygen and chlorine.

Which mass of this product is formed?

A 21.3 g **B** 44.7 g **C** 63.9 g **D** 128 g

18 The following two experiments are carried out with anhydrous potassium chloride and observations X and Y are made at the end of each experiment.

Concentrated sulfuric acid is added to the potassium chloride and the fumes produced are bubbled into aqueous potassium iodide solution - observation X.

The potassium chloride is dissolved in aqueous ammonia and this is then added to aqueous silver nitrate - observation Y.

What are the observations X and Y?

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16 Aqueous sodium chloride (brine) is electrolysed by using inert electrodes in a cell which is stirred so that products of electrolysis react with each other. The cell is kept cold.

Which pair of substances is among the major products?

A hydrogen and chlorine

B hydrogen and sodium chlorate(I)

C hydrogen and sodium chlorate(V)

D sodium hydroxide and chlorine

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A as an acid only

B as an acid and oxidising agent

C as an oxidising agent only

D as a reducing agent only

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Answer Key

No.	Paper	Topic	Answer
1	9701_m25_qp_12	11. Group 17	**B**
2	9701_s24_qp_11	11. Group 17	**C**
3	9701_w24_qp_11	11. Group 17	**C**
4	9701_w24_qp_13	11. Group 17	**C**
5	9701_m22_qp_12	11. Group 17	**B**
6	9701_s22_qp_13	11. Group 17	**A**
7	9701_w21_qp_13	11. Group 17	**D**
8	9701_w20_qp_12	11. Group 17	**B**
9	9701_s19_qp_11	11. Group 17	**D**
10	9701_m17_qp_12	11. Group 17	**A**
11	9701_m16_qp_12	11. Group 17	**A**
12	9701_s16_qp_12	11. Group 17	**A**
13	9701_s16_qp_13	11. Group 17	**D**
14	9701_s15_qp_13	11. Group 17	**C**
15	9701_s14_qp_13	11. Group 17	**A**
16	9701_s12_qp_11	11. Group 17	**C**
17	9701_s12_qp_13	11. Group 17	**C**
18	9701_w11_qp_11	11. Group 17	**B**
19	9701_w11_qp_13	11. Group 17	**B**
20	9701_s10_qp_11	11. Group 17	**A**
21	9701_s10_qp_12	11. Group 17	**A**
22	9701_s10_qp_13	11. Group 17	**A**